Expt. No. 6

Title: To design layout of SR flip-flop using Micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

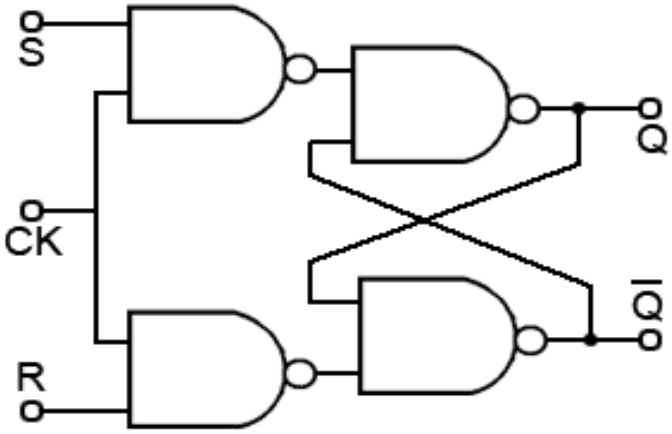
Experiment No. 6

Aim: To design layout of SR flip-flop using DSCH and Micro-wind.

Software Required: DSCH, Micro-wind software

Hardware: computer

Circuit Diagram:



Theory:

SR flip-flop can be implemented using cross NAND structure. SR flip- flop shown in which does not match with synchronous. A hacked version of the latch is shown. It consist of CMOS inverter of transistor to device flip-flop from one state to another private synchronous control is steady state where o the one is slow.

1. N-well/p-well spacing = 6A
2. N-well p-well width = 6A
3. Active spacing = 3A
4. Metal width = 2A
5. Metal to metal = 3A
6. Poly width = 3A
7. Mask = 3A

Procedure:

1] Start micro-wind software for layout**.**

2] Draw the SR flip-flop following the design.

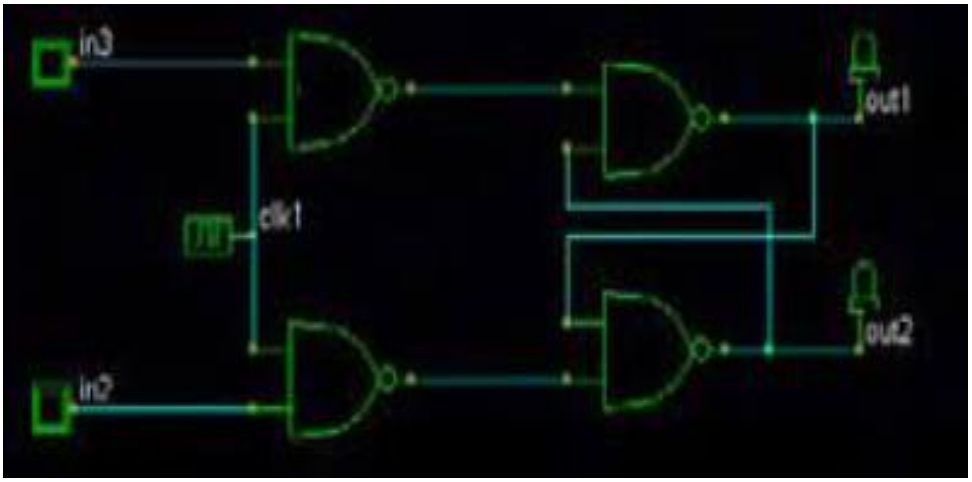
3] Click on DRC at every stage of design to avoid error.

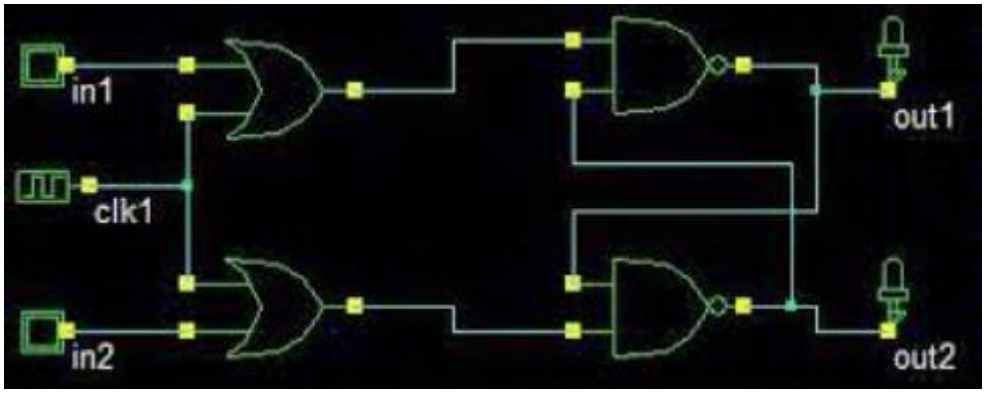
4] Observe various output while using simulation output.

5] Software design and finally run a DRC.

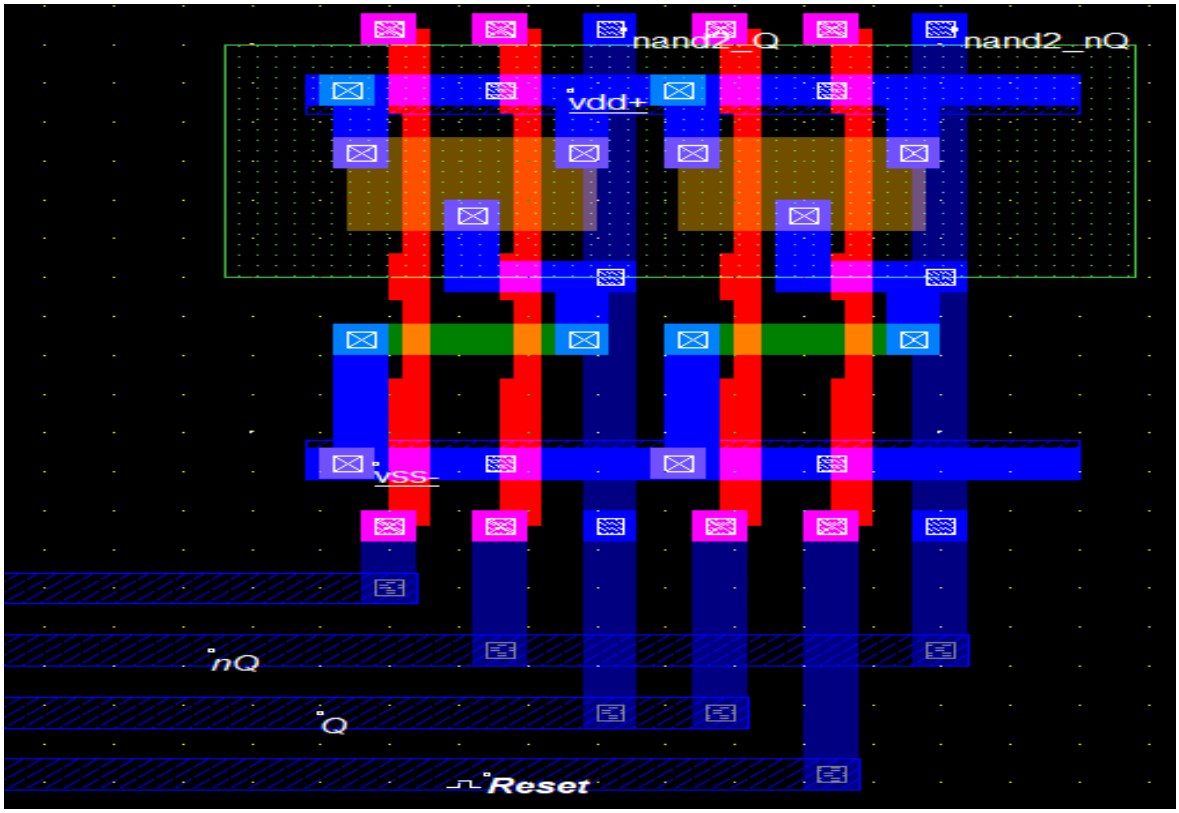
6] Record the output waveform observed.

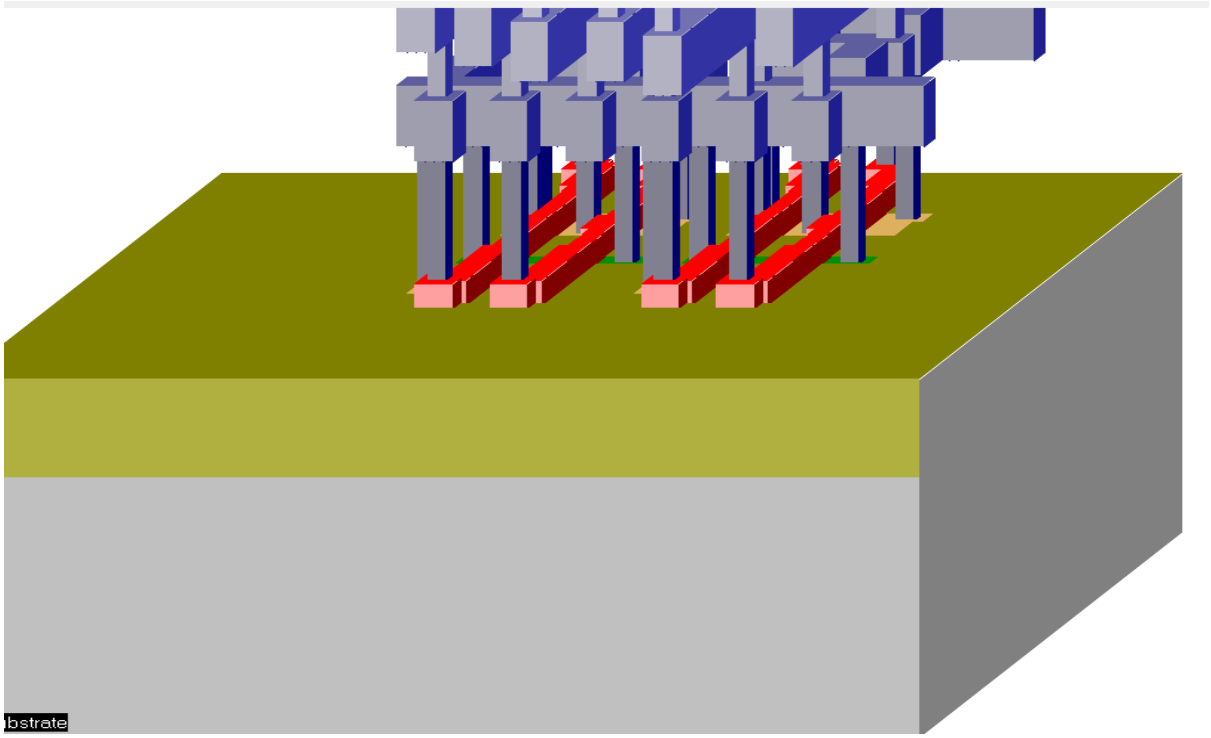
DSCH:


in3
clk1
in2
out1
out2


in1
clk1
in2
out1
out2

Layout:


nand2_Q
nand2_nQ
vdd+
vss-
nQ
Q
Reset



Pre Lab-Questions:

1. Why S-R flip-flop can be constructed by cross-coupling?
2. Why S-R latch composed of NAND gates?

Post Lab Questions:

1 Why Is the implant in PMOS?
2 Why metal 1 and metal 2 layer Indicate?

Conclusion: Thus, we have designed layout of SR flip flop and verified layout design

CO's covered:

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